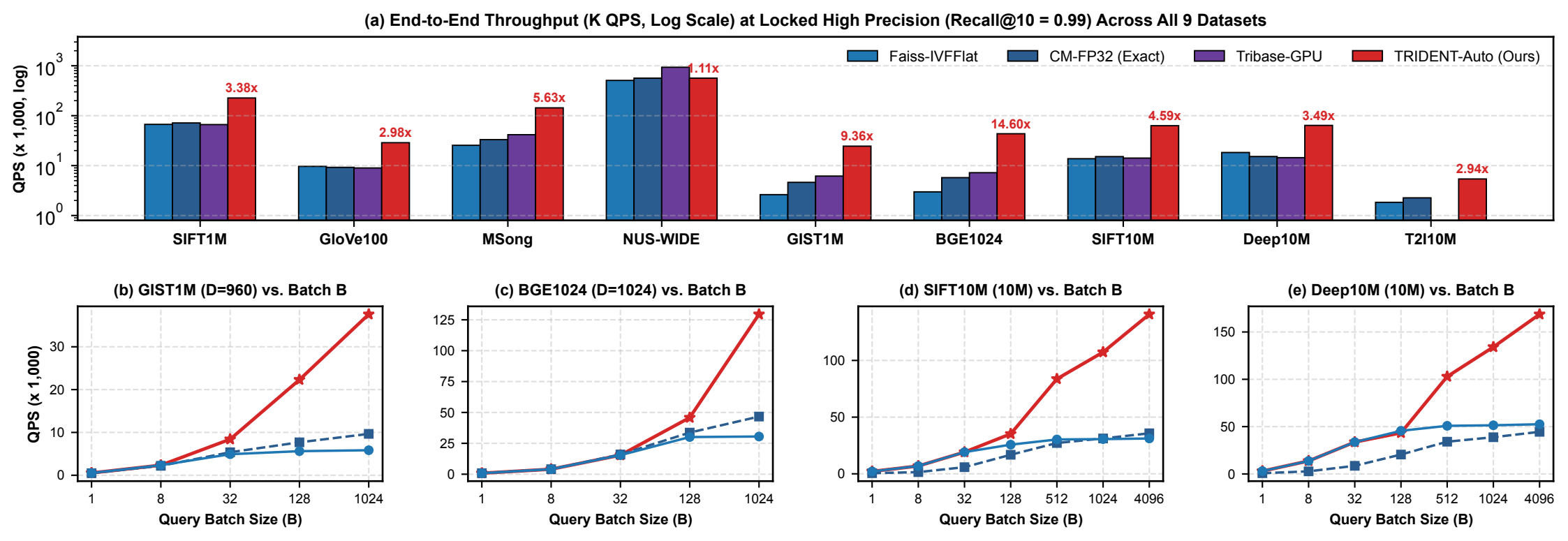
(a) End-to-End Throughput (K QPS, Log Scale) at Locked High Precision (Recall@10 = 0.99) Across All 9 Datasets
QPS (x 1,000, log)
Faiss-IVFFlat
CM-FP32 (Exact)
Tribase-GPU
TRIDENT-Auto (Ours)
3.38x
2.98x
5.63x
1.11x
9.36x
14.60x
4.59x
3.49x
2.94x
SIFT1M
GloVe100
MSong
NUS-WIDE
GIST1M
BGE1024
SIFT10M
Deep10M
T2I10M
(b) GIST1M (D=960) vs. Batch B
(c) BGE1024 (D=1024) vs. Batch B
(d) SIFT10M (10M) vs. Batch B
(e) Deep10M (10M) vs. Batch B
QPS (x 1,000)
Query Batch Size (B)